

Chapter 1. Sequential Circuits

Excitation tables of flip-flops

Analysis of sequential circuits

Implementation of sequential circuits from given state tables

Excitation Tables of Flip-Flops

Excitation table shows what input is needed for transitions of flip-flops' states.

Excitation of D flip-flop:

In the excitation table below, q represents present output and Q represents the next output.

What this table tells us is;

- If we want a D flip-flop's output to keep its value as 0, we need to input 0.
- If we want a D flip-flop's output to go from 0 to 1, we need to input 1.
- If we want a D flip-flop's output to go from 1 to 0, we need to input 0.
- If we want a D flip-flop's output to keep its value as 1, we need to input 1.

q	Q	D
0	0	0
0	1	1
1	0	0
1	1	1

Excitation Tables of Flip-Flops

Excitation table of SR flip-flop:

S	R	q	Q
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	-
1	1	1	-

q	Q	S	R
0	0	0	x
0	1	1	0
1	0	0	1
1	1	x	0



For this transition, both inputs given below does the same thing.

SR = 00

SR = 01

As long as S=0, we **don't care** about R.
So, SR is written as 0x.

Excitation Tables of Flip-Flops

Excitation table of T flip-flop:

q	Q	T
0	0	0
0	1	1
1	0	1
1	1	0

Excitation table of JK flip-flop:

q	Q	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

$\Rightarrow JK=00$ or $JK=01$
 $\Rightarrow JK=10$ or $JK=11$
 $\Rightarrow JK=01$ or $JK=11$
 $\Rightarrow JK=00$ or $JK=10$

Analysis of Sequential Circuits

Analysis of sequential circuits require 3 steps:

1. Deriving equations of outputs and next states of the flip-flops.
2. Generating state tables which shows the input, output and the next states. This table shows what data to store in the memory components when the next clock pulse is received.
3. Generating state diagrams from state tables.

Analysis of Sequential Circuits

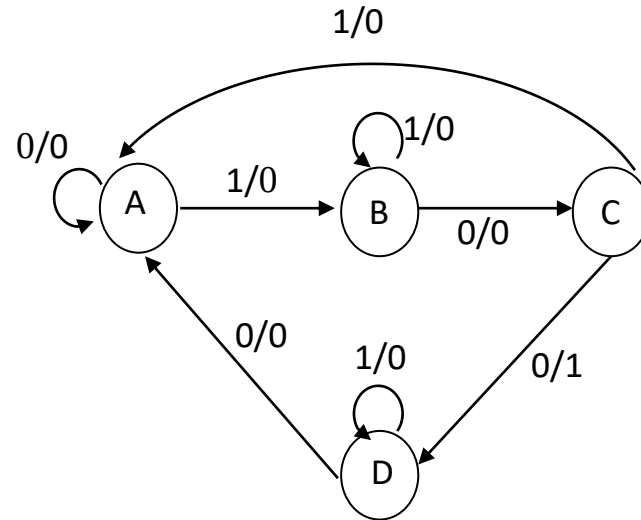
Let's start with basic terms state, state table and state diagram.

State of a circuit means the outputs of flip-flops used. Generally, letters are used as nicknames for the states. (i.e. A = 00 in a circuit with 2 flip-flops)

Present State	Next State		Output (z)	
	x=0	x=1		
A	A	B	0	0
B	C	B	0	0
C	D	A	1	0
D	A	D	0	0

State Table

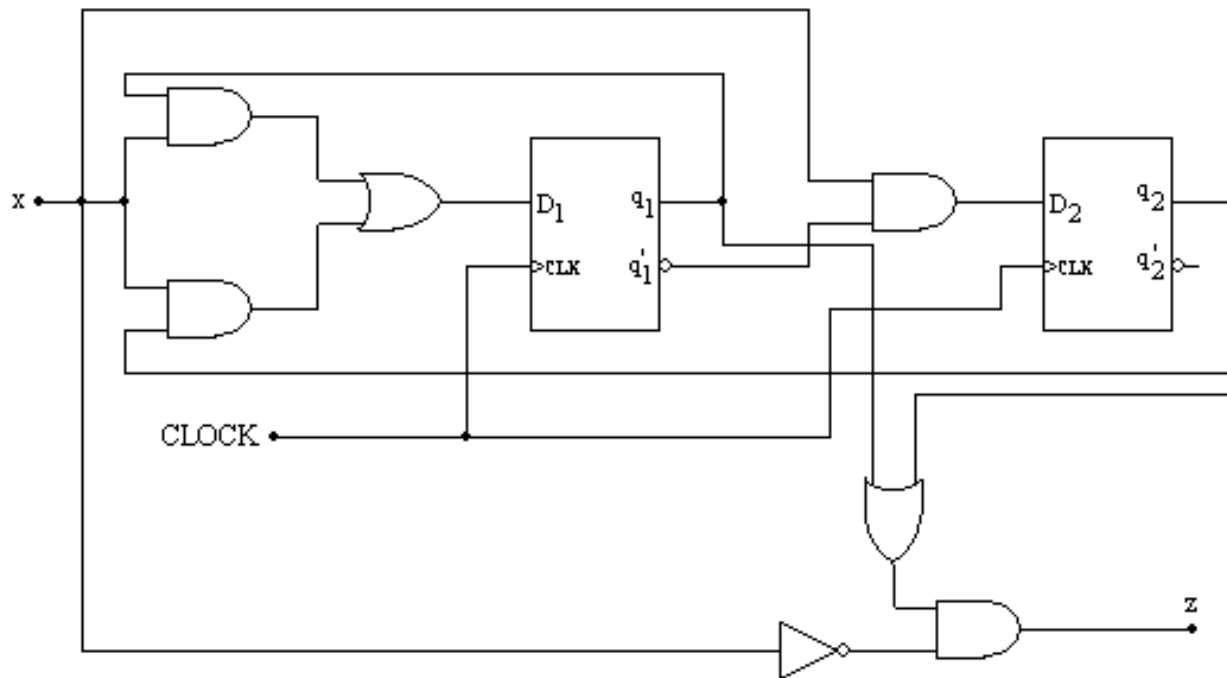
↑
States



State Diagram

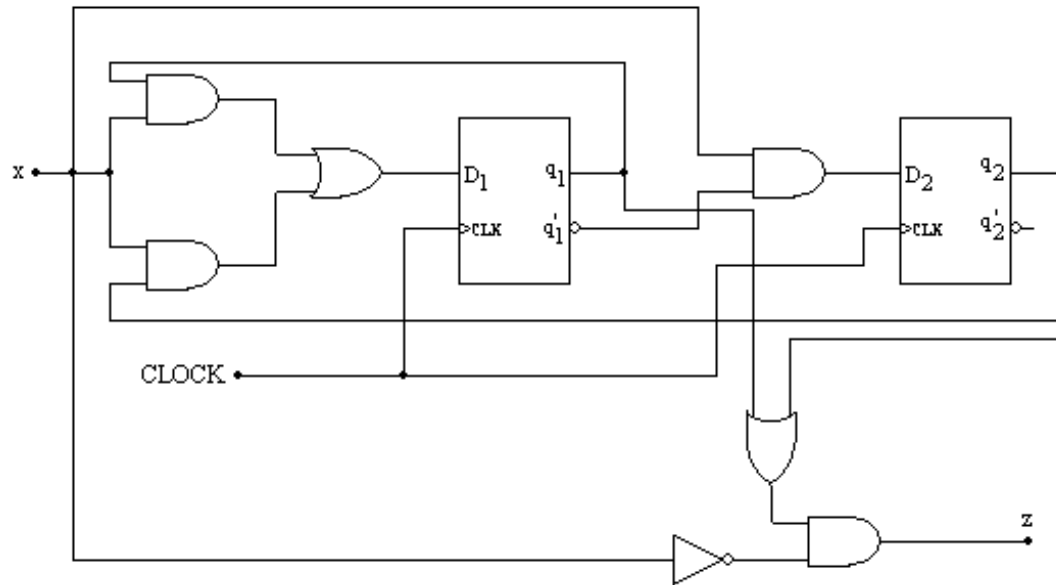
Analysis of Sequential Circuits

Example:



In the given circuit, we can tell what the input is, what the output is and how many memory components are used.

In this circuit, there is a single input and a single output. And there are 2 flip-flops used as memory components.



$$D_1 = x.q_1 + x.q_2 = x.(q_1 + q_2)$$

$$D_2 = x.q_1,$$

So, $Q_1 = x.(q_1+q_2)$ and $Q_2 = x.q_1$,

And the output $z = x' \cdot (q_1 + q_2)$

Example (3rd Page):

In the second step, we need to generate state tables using the state equations derived in the first step.

$$Q_1 = x.(q_1 + q_2)$$

$$Q_2 = x.q_1'$$

$$z = x'.(q_1 + q_2)$$

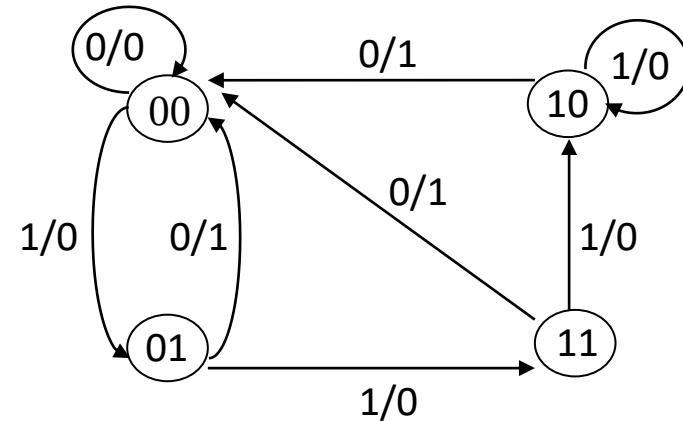
Present State			Next State		Output
q_1	q_2	x	Q_1	Q_2	z
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

Present State ($q_1 q_2$)	Next State ($Q_1 Q_2$)		Output (z)	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
(A) 0 0	0 0	0 1	0	0
(B) 0 1	0 0	1 1	1	0
(C) 1 0	0 0	1 0	1	0
(D) 1 1	0 0	1 0	1	0

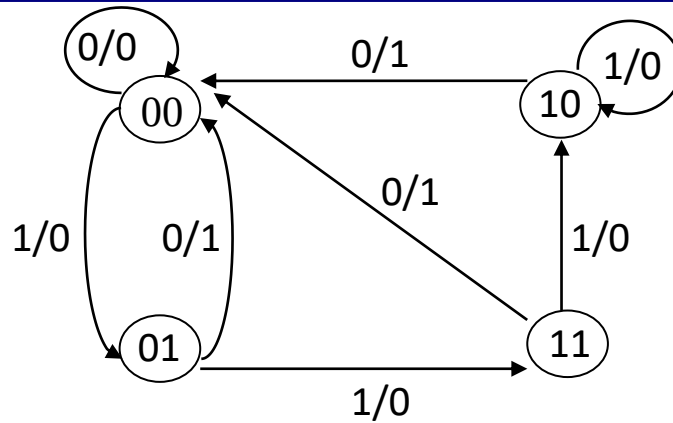
Example (4th Page):

Last step is to generate the state diagram.

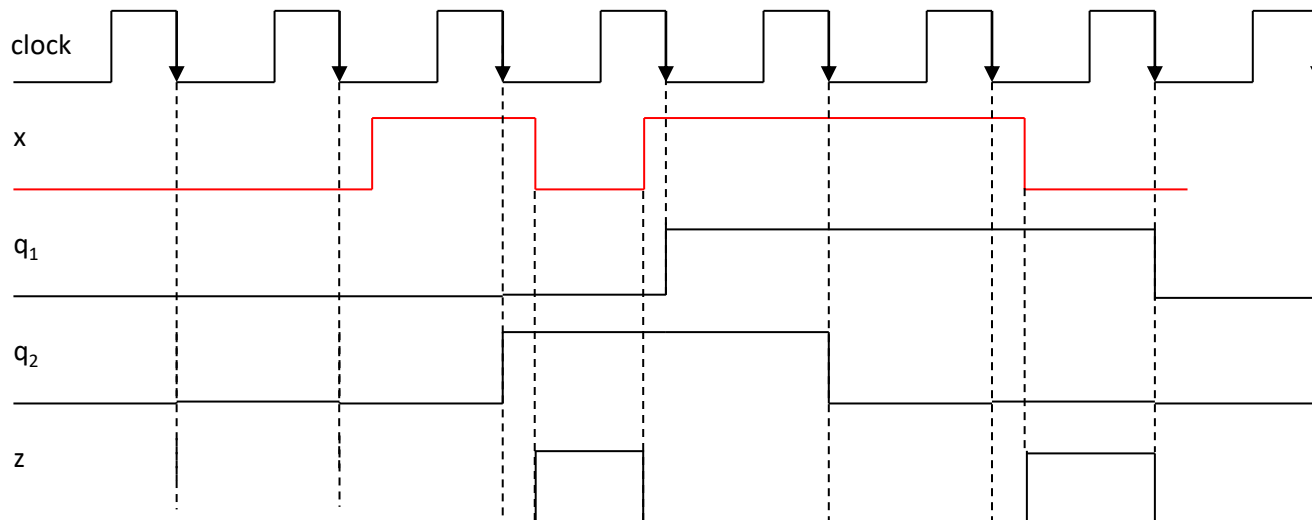
Present State ($q_1 q_2$)	Next State ($Q_1 Q_2$)		Output (z)	
	$x = 0$	$x = 1$	$x = 0$	$x = 1$
(A) 0 0	0 0	0 1	0	0
(B) 0 1	0 0	1 1	1	0
(C) 1 0	0 0	1 0	1	0
(D) 1 1	0 0	1 0	1	0



Example (5th Page):



Let's analyze this circuit's behavior on the timing diagram below.



Implementation of Sequential Circuits from Given State Tables

Example: Let's implement a sequential circuit from the state table below.

Present State	Next State		Output (z)	
	x=0	x=1	x=0	x=1
A	A	B	0	0
B	C	B	0	0
C	D	A	1	0
D	A	D	0	0

State table has 4 states. This means that we need to use 2 flip-flops.

We can name the states as A, B, C and D.

States	$y_1 y_2$
A	0 0
B	0 1
C	1 0
D	1 1

or

States	$y_1 y_2$
A	0 0
B	0 1
C	1 1
D	1 0

or we can name states in some other way.

Example (2nd Page):

States	$y_1 y_2$
A	0 0
B	0 1
C	1 0
D	1 1



Present State	Next State		Output (z)	
	x=0	x=1	x=0	x=1
A	A	B	0	0
B	C	B	0	0
C	D	A	1	0
D	A	D	0	0



Present State ($y_1 y_2$)	Next State ($Y_1 Y_2$)		Output (z)	
	x=0	x=1	x=0	x=1
0 0	0 0	0 1	0	0
0 1	1 0	0 1	0	0
1 0	1 1	0 0	1	0
1 1	0 0	1 1	0	0

Example (3rd Page):

State Table

Present State ($y_1 y_2$)	Next State ($Y_1 Y_2$)		Output (z)	
	$x=0$	$x=1$	$x=0$	$x=1$
0 0	0 0	0 1	0	0
0 1	1 0	0 1	0	0
1 0	1 1	0 0	1	0
1 1	0 0	1 1	0	0



Truth Table

Present State $x y_1 y_2$	Next State $Y_1 Y_2$	Inputs of Flip-Flops				Output z
		J_1	K_1	J_2	K_2	
0 0 0	0 0	0	x	0	x	0
0 0 1	1 0	1	x	x	1	0
0 1 0	1 1	x	0	1	x	1
0 1 1	0 0	x	1	x	1	0
1 0 0	0 1	0	x	1	x	0
1 0 1	0 1	0	x	x	0	0
1 1 0	0 0	x	1	0	x	0
1 1 1	1 1	x	0	x	0	0

q	Q	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0



Example (4th Page):

Present State $x y_1 y_2$	Next State $Y_1 Y_2$	Inputs of Flip-Flops $J_1 K_1 J_2 K_2$	Output z
0 0 0	0 0	0 x 0 x	0
0 0 1	1 0	1 x x 1	0
0 1 0	1 1	x 0 1 x	1
0 1 1	0 0	x 1 x 1	0
1 0 0	0 1	0 x 1 x	0
1 0 1	0 1	0 x x 0	0
1 1 0	0 0	x 1 0 x	0
1 1 1	1 1	x 0 x 0	0

J_1				
$y_1 y_2$	00	01	11	10
x				
0	0	1	x	x
1	0	0	x	x

$$J_1 = x'y_2$$

K_1				
$y_1 y_2$	00	01	11	10
x				
0	x	x	1	0
1	x	x	0	1

$$K_1 = xy_2' + x'y_2 = x \oplus y_2$$

J_2				
$y_1 y_2$	00	01	11	10
x				
0	0	x	x	1
1	1	x	x	0

$$J_2 = xy_1' + x'y_1 = x \oplus y_1$$

K_2				
$y_1 y_2$	00	01	11	10
x				
0	x	1	1	x
1	x	0	0	x

$$K_2 = x'$$

From the truth table;

$$z = x'y_1y_2'$$

Example (5th Page):

$$J_1 = x'y_2$$

$$K_1 = xy_2' + x'y_2 = x \oplus y_2$$

$$J_2 = xy_1' + x'y_1 = x \oplus y_1$$

$$K_2 = x'$$

